

Econ 440: Experimental Design: Application to the Evaluation of the Job Corps Training Program

Carlos Flores

Department of Economics
California Polytechnic State University

1 The Job Corps Training Program: Background

Please see slides in Canvas.

2 The National Job Corps Study (NJCS)

- In the 1990's, there was a randomized social experiment to evaluate the effectiveness of the Job Corps (JC) training program.
- Design:

Design of the NJCS

First-time Applicants between
1994-1995 in the 48 contiguous states



Check Eligibility



Get "Target" Population
(our population of interest)



Part of the
NJCS

} First randomization:
Get a random
sample of
eligible applicants
(our population)

} Second randomization:
RA of T

Not part
of the NJCS

T=1: Allowed to
Enroll in JC

T=0: Not allowed
to enroll in JC for
3 years

- Surveys were given at baseline (i.e., at RA), and at 52, 130, and 208 weeks after RA.
- Note: Due to implementation and design issues, females were assigned to treatment ($T = 1$) and control ($T = 0$) groups with different probabilities than males, i.e., $Pr(T = 1|female) \neq Pr(T = 1|males) \implies$ “gender” is not balanced between treatment and control groups.
 - *Why?* Some JC centers had mostly men $\implies$ Let $Pr(T = 1|female)$ be higher to ensure we had females in every center in our sample.
 - *Options to deal with imbalance:*
 - * Ignore issue (not good practice, as “gender” is not balanced)
 - * Do analysis separately for males and females
 - * Use “sampling weights” in the NJCS (not for this class)

3 Econometric Analysis

The rest of these notes provide a “sketch” of the econometric analysis. For details, please see the corresponding lecture.

3.1 Exploring the Data

- Open the data “Example_Exper_Job_Corps_Full.dta” (available in Canvas). Type “describe” or “des” to see a description of the data.
- Key variables:
 - Treatment: “treatmnt”
 - Outcomes: Earnings (*earnq16*) and employment (*workq16*) in quarter 16 (i.e., 4 years) after RA
 - Covariates: Rest of the variables in the data set.

Goal: Estimate effect of Job Corps on earnings and employment 4 years after RA.

- *Some Stata Commands:*
 - Get the average of all variables by using: “sum” or “mean _all.”

$\implies$ on average, individuals are: 18 years old, 10 years of education, only 1 in 4 have finished high school, 1 in 5 have a child, 60% are unemployed, 13% have been found guilty of a crime, about half are black, etc.

$\implies$ Economically disadvantaged kids.

- Look at the average of our outcomes.
- Look at how many treated and control individuals we have by tabulating the treatment variable: “tab treatmnt”.

3.2 Did Randomization Work in Balancing Covariates?

- Get means by treatment status. There are a few ways to do this in Stata:
 - “sum _all if treat==1” and “sum _all if treat==0”
 - “mean _all if treat ==1” and “mean _all if treat ==0”
 - To see the means next to each other: “mean [*list of variables here*] , over(treat)”

Note: Remember the difference between standard deviation (reported by the “sum” command) and standard error (reported by the “mean” command):

Standard Deviation of X_i : $SD(X_i) = \sqrt{var(X_i)}$

Standard Error of $\bar{X}$: $\sqrt{\frac{var(X_i)}{n}} = \frac{\sqrt{var(X_i)}}{\sqrt{n}}$

We use “standard deviation” for the observations X_i , and “standard error” for $\bar{X}$ (the sample mean, an estimator).

- How to perform a test of equality of means in Stata (or other programs)?
There are at least three ways:

1. Use the “means” command and then manually calculate the t-statistic. Let $\hat{E}(X|T=1) = \bar{X}_1$ and $\hat{E}(X|T=0) = \bar{X}_0$. Then:

$$t - stat = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{Var(\bar{X}_1 - \bar{X}_0)}} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}}}$$

where $\frac{\hat{\sigma}_0^2}{n_0}$ is the standard error of $\bar{X}_0$ and $\frac{\hat{\sigma}_1^2}{n_1}$ is the standard error of $\bar{X}_1$.

2. In Stata, use the “ttest” command: “ttest age, by(treat)” (to check if age is balanced between the treatment vs. control group).

3. Use OLS. In Stata: “reg age treat”, where the coefficient for T gives the difference in the sample average of age between the treatment and control group (to check if age is balanced across the treatment vs. control group).

You can replace “age” with any of the other covariates to test.

- Test a few, as examples:
 - black
 - baseline earnings (basewee) and employment (emp)
 - guilt
 - education: hgcb, gs-ged
 - age. They are statistically different, but the means are very close. We have a large sample (almost 10,000 observations), so we are more likely to reject the null even for small differences in means. However, note that the difference in means is very small (18.75 years versus 18.88). This difference is unlikely to have a big impact for us in terms of affecting our results (say, a difference of a few months in age is unlikely to have a big impacts on earnings). Thus, while the difference is statistically significant, it is likely “*economically insignificant*”.
 - females: as expected; large difference and statistically significant.
 - Child: reject, likely because of the imbalance in “female”. Note:
 - * the “mean child if female == 1” is 0.325 (1/3 of females have kid)
 - * the “mean child if female == 0” is 0.10 (1/10 of males have kid)

$\implies$ If we restrict to males and females (e.g., “reg child treat if female == 1” and “reg child treat if female == 0”) $\implies$ Child is balanced

- Overall, the balance in the NJCS looks good.

3.3 Estimate the Average Treatment Effect

- For simplicity, we estimate the effect of Job Corps on our outcomes ignoring the fact that the “female” indicator is not balanced (remember, that variable is not balanced “by design” in this experiment, as previously discussed).
 - Estimate of the ATE on earnings: **\$14 per week**; statistically significant.
Interpretation: The average effect of the Job Corps training program on weekly earnings four years after RA for eligible individuals (our population) is an increase of \$14 per week.

- Estimate of the ATE on employment: **0.021**; statistically significant. Interpretation: The average effect of the Job Corps training program on the probability of being employed four years after RA for eligible individuals (our population) is an increase of 0.021.

- Is the effect big enough to make a difference? In other words, is the estimated effect of Job Corps economically significant? It does not matter if the effect is statistically significant if it is so small (e.g., a penny) to not make a difference in the life of these individuals.

- To get a sense about the size or magnitude of the effect, it is always good to look at it in percentage terms relative to the expected value have these individuals not enrolled in the program, $E[Y(0)] = E[Y|T = 0]$, by RA of T . In other words, we can look at:

$$\frac{ATE}{E(Y|T = 0)}$$

- In our case:

* Earnings: $\frac{14}{200} = 7\%$. Or, $14 \cdot 52 = \$728$ per year.

* Employment: $\frac{0.022}{0.693} = 3\%$.

$\implies$ The effects of Job Corps are **both** *economically and statistically* significant, especially on earnings.

- Analyze the effect of adding covariates to our OLS of the outcome on the treatment.
 - Adding all but “female” (which is not balanced by design): results are about the same as before (as expected).
 - Even if we add “female”, results are about the same as before.

$\implies$ Adding covariates to the OLS does not change our estimate of the ATE much.

- Since “female” is not balanced, it is more appropriate to estimate the effects separately by gender. Doing this, we find (without adding covariates):

For Males:

- Estimate of the *ATE* on earnings: \$22.90 (t-stat: 3.90)
- Estimate of the *ATE* on employment: 0.019 (1.63)

For Females:

- Estimate of the *ATE* on earnings: \$13.73 (2.72)
- Estimate of the *ATE* on employment: 0.032 (2.42)
- As before, results do not change much if we add covariates to the OLS.
- Note: Effects on earnings are larger for males (almost double!), while effects on employment are larger for females.

- Treatment Effect Heterogeneity

This refers to how the effect of interest varies for different subpopulations or groups of interest. Above, we did this by gender. It can also be analyzed for other groups. For example, effect on earnings by race/demographic groups:

- White: \$28.20 (t-stat: 3.24)
- Black: \$19.60 (3.51)
- Hispanics: -\$18.94 (-1.85)

Remember: The *ATE* is an *average* effect, it does not mean that the effect of Job Corps on every individual in the population is the same and it equals to the *ATE* (say, the effect of JC on earnings is not \$14 per week for *every* individual!). Indeed, in practice, treatment effect heterogeneity is very important. Usually, treatment effects differ across individuals and over baseline characteristics (e.g., by gender, race, age, etc.). The *ATE* is only that, and *average*.